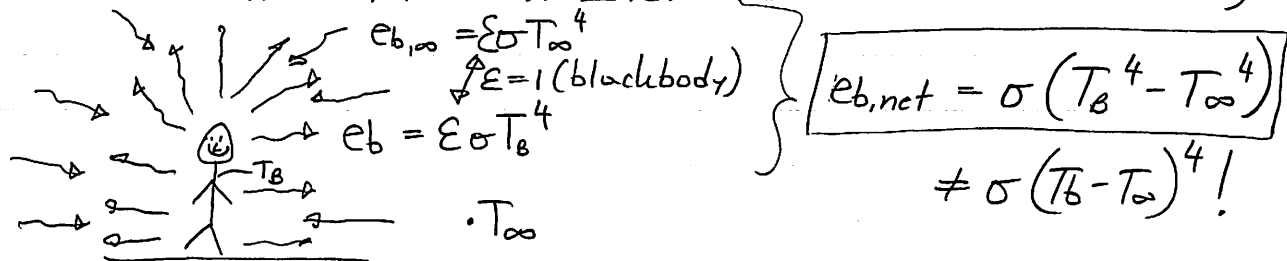
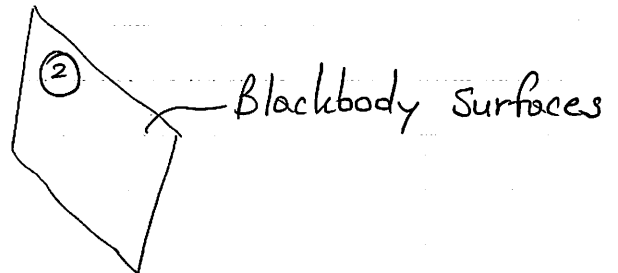
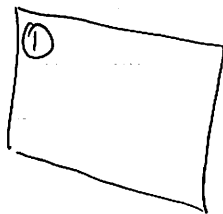


View Factors and Blackbody Heat Exchange

So far, everything has been about blackbody radiation to an infinite medium. I.e.: (surfaces see each other)



But, what if we have finite surfaces:



$$\left. \begin{aligned} q_{1 \rightarrow 2} &= F_{12} A_1 e_{b1} \\ q_{2 \rightarrow 1} &= F_{21} A_2 e_{b2} \end{aligned} \right\} \begin{aligned} F_{ab} &\equiv \text{view factor} \\ &\equiv \text{fraction of radiation leaving surface "a" and reaching "b"} \end{aligned}$$

So our net exchange is:

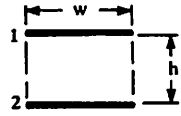
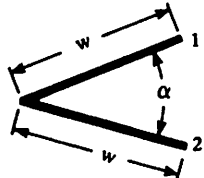
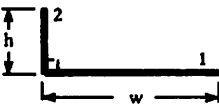
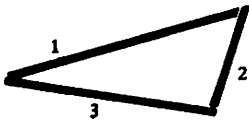
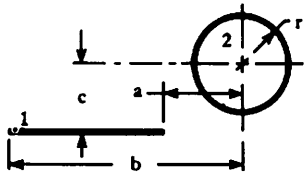
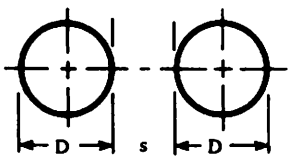
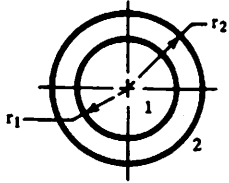
$$q_{12} = q_{1 \rightarrow 2} - q_{2 \rightarrow 1} = F_{12} A_1 e_{b1} - F_{21} A_2 e_{b2}$$

But if $T_1 = T_2$, $e_{b1} = e_{b2}$

$$q_{12} = e_b (A_1 F_{12} - A_2 F_{21}) = 0 \quad (\text{no net energy transfer in, between surfaces at same } T)$$

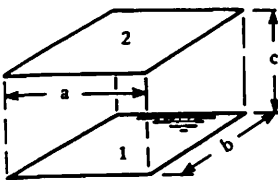
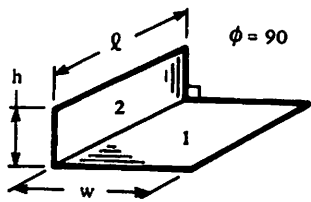
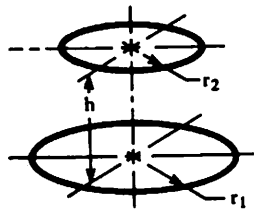
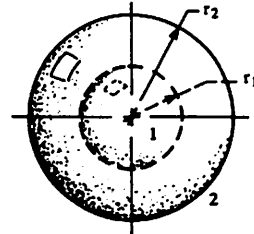
$$\boxed{A_1 F_{12} = A_2 F_{21}} \Rightarrow \text{Reciprocity}$$

Table 10.2 View factors for a variety of two-dimensional configurations (infinite in extent normal to the paper)

Configuration	Equation
1. 	$F_{1-2} = F_{2-1} = \sqrt{1 + \left(\frac{h}{w}\right)^2} - \left(\frac{h}{w}\right)$
2. 	$F_{1-2} = F_{2-1} = 1 - \sin(\alpha/2)$
3. 	$F_{1-2} = \frac{1}{2} \left[1 + \frac{h}{w} - \sqrt{1 + \left(\frac{h}{w}\right)^2} \right]$
4. 	$F_{1-2} = (A_1 + A_2 - A_3)/2A_1$
5. 	$F_{1-2} = \frac{r}{b-a} \left[\tan^{-1} \frac{b}{c} - \tan^{-1} \frac{a}{c} \right]$
6. 	Let $X = 1 + s/D$. Then: $F_{1-2} = F_{2-1} = \frac{1}{\pi} \left[\sqrt{X^2 - 1} + \sin^{-1} \frac{1}{X} - X \right]$
7. 	$F_{1-2} = 1$, $F_{2-1} = \frac{r_1}{r_2}$, and $F_{2-2} = 1 - F_{2-1} = 1 - \frac{r_1}{r_2}$

Adapted From Lienhard & Lienhard "A Heat Transfer Textbook" (2012)

Table 10.3 View factors for some three-dimensional configurations

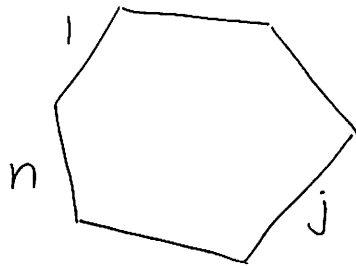
Configuration	Equation
1. 	Let $X = a/c$ and $Y = b/c$. Then: $F_{1-2} = \frac{2}{\pi XY} \left\{ \ln \left[\frac{(1 + X^2)(1 + Y^2)}{1 + X^2 + Y^2} \right]^{1/2} - X \tan^{-1} X - Y \tan^{-1} Y + X \sqrt{1 + Y^2} \tan^{-1} \frac{X}{\sqrt{1 + Y^2}} + Y \sqrt{1 + X^2} \tan^{-1} \frac{Y}{\sqrt{1 + X^2}} \right\}$
2. 	Let $H = h/\ell$ and $W = w/\ell$. Then: $F_{1-2} = \frac{1}{\pi W} \left\{ W \tan^{-1} \frac{1}{W} - \sqrt{H^2 + W^2} \tan^{-1} (H^2 + W^2)^{-1/2} + H \tan^{-1} \frac{1}{H} + \frac{1}{4} \ln \left[\frac{(1 + W^2)(1 + H^2)}{1 + W^2 + H^2} \right] \times \left[\frac{W^2(1 + W^2 + H^2)}{(1 + W^2)(W^2 + H^2)} \right]^{W^2} \left[\frac{H^2(1 + H^2 + W^2)}{(1 + H^2)(H^2 + W^2)} \right]^{H^2} \right\}$
3. 	Let $R_1 = r_1/h$, $R_2 = r_2/h$, and $X = 1 + (1 + R_2^2)/R_1^2$. Then: $F_{1-2} = \frac{1}{2} \left[X - \sqrt{X^2 - 4(R_2/R_1)^2} \right]$
4. 	Concentric spheres: $F_{1-2} = 1, \quad F_{2-1} = (r_1/r_2)^2, \quad F_{2-2} = 1 - (r_1/r_2)^2$

Now we can formulate our analysis as:

$$q_{12} = \underbrace{A_1 F_{12} (e_{b1} - e_{b2})}_{A_1 F_{12} e_{b1} - A_2 F_{21} e_{b2}} \Leftrightarrow$$

$\underbrace{A_1 F_{12}}_{\text{Radiative resistance}}$

For enclosures:

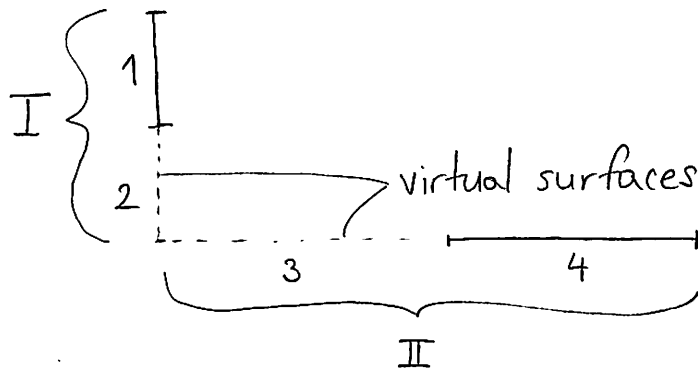


$$\boxed{\sum_{j=2}^n F_{ij} = 1} \Rightarrow \text{True for flat or convex surfaces.}$$

If concave surface 

$$\sum_{j=2}^n F_{ij} \neq 1 \quad \text{since } F_{ii} \neq 0 \quad (\text{Surface sees itself})$$

Example | Consider the following arrangement between two perpendicular plates 1 & 4



$$A_1 F_{14} = A_I F_{I4} - A_2 F_{24}$$

$$A_I^I F_{I4} = A_I^I F_{I\text{II}} - A_I^{\text{II}} F_{I3} \Rightarrow F_{I4} = F_{I\text{II}} - F_{I3}$$

$$A_2 F_{24} = A_2 F_{2II} - A_2 F_{23} \Rightarrow F_{24} = F_{2II} - F_{23}$$

Can look these up
in view factor
tables.